Corrected SQR Optimization with a Fock-Resolved Effective-Qubit Metric

OpenAI Codex
Autonomous cQED Study Workspace
(Dated: March 27, 2026)

We revisited direct Gaussian multitone SQR optimization after the recent convention fixes in `cqed_sim`. The main source-level check was the phase convention: in the current checkout, the reduced conditioned-multitone layer now treats ϕ_n as the same equatorial rotation-axis angle used by $R_\phi(\theta)$, not as a Bloch-azimuth proxy with a hidden quarter-turn shift. With that correction confirmed, we reran the study using a smooth four-level Fock-resolved target profile and compared three reduced objectives: the baseline corrected waveform, optimization of the reduced final-state fidelity, and optimization of a stricter reduced effective-unitary process-fidelity metric. The waveform family was the corrected direct Gaussian multitone ansatz with per-tone amplitude, phase, and carrier corrections. A small-angle kernel analysis showed no first-order reachability obstruction: the reduced multitone kernel remained full rank throughout the scan, although its condition number worsened as the addressed Fock window grew or the pulse shortened. Numerically, the reduced effective-unitary metric remained the right optimization target. The strongest nontrivial four-level point reached weighted effective-unitary fidelity 0.999978 at $N_{\text{active}} = 4$ and $|\chi|T/2\pi = 3.0$, improving on a baseline value of 0.992316. The clearest gain appeared in the hardest short-duration case, where $N_{\text{active}} = 4$ and $|\chi|T/2\pi = 1.0$ improved from 0.793061 to 0.961990. State-based optimization remained misleading even after the convention fix: it could increase final-state fidelity while substantially worsening the effective-unitary metric. The corrected direct multitone family therefore appears expressive enough for this smooth corrected-SQR target at moderate durations, while the reduced effective-unitary metric remains essential for honest assessment.

I. INTRODUCTION

Selective qubit rotation (SQR) in a dispersive qubit-cavity system is most naturally viewed as a Fock-resolved qubit-rotation profile. The user asked for a reduced optimization criterion that ignores detailed cavity-subspace structure but still checks whether each relevant Fock sector induces the intended effective qubit rotation parameters (θ_n, ϕ_n). This report first verifies the corrected convention now used by `cqed_sim`, then derives a small-angle reachability picture, and finally compares reduced state-based and reduced effective-unitary metrics for the corrected direct multitone waveform.

II. SYSTEM AND METHODS

A. Hamiltonian

The reduced sector model uses

$$H_n(t) = \Delta_n |e\rangle\langle e| + \epsilon(t)\sigma_+ + \epsilon^*(t)\sigma_-, \tag{1}$$

where

$$\epsilon(t) = g(t/T) \sum_m A_m e^{i[(\omega_m+\delta\omega_m)t + (\phi_m+\delta\alpha_m)]}, \tag{2}$$

with normalized Gaussian envelope g and corrected amplitude convention

$$A_n = \frac{\theta_n}{2T} + \lambda_0 \delta\lambda_n, \qquad \lambda_0 = \frac{\pi}{2T}. \tag{3}$$

B. Simulation Parameters

TABLE I. Core simulation parameters.

Parameter	Value
$\omega_q/2\pi$	6.150 GHz
$\omega_c/2\pi$	5.241 GHz
$\alpha/2\pi$	-255 MHz
$\chi/2\pi$	-2.84 MHz
n_{cav}	8
n_{tr}	2
dt	4 ns
σ/T	1/6
$\lvert\chi\rvert T/2\pi$ sweep	1, 3, 5
N_{active} sweep	1, 2, 3, 4

C. Analytic Preliminary

At small angles, the excited-state amplitude in sector n obeys

$$c_{e,n} \approx -iT \sum_m K_{nm} a_m, \tag{4}$$

with

$$K_{nm} = \int_0^1 g(x) e^{i(\Delta_n - \Delta_m)Tx}\, dx. \tag{5}$$

The current source now uses the same ϕ_n convention in the reduced layer and in $R_\phi(\theta)$, so the corrected target state from $|g\rangle$ is

$$|\psi_n\rangle = \cos\frac{\theta_n}{2}|g\rangle - ie^{i\phi_n} \sin\frac{\theta_n}{2}|e\rangle. \tag{6}$$

The kernel remained full rank in every scanned case, showing that low-angle reachability exists in principle. The difficulty is quantitative: the kernel becomes more ill conditioned as the active window grows or the gate shortens, so the naive direct prescription $A_n = \theta_n/(2T)$ increasingly departs from the first-order inverse solution.

D. Computational Approach

Two reduced metrics were compared. The first used `cqed_sim.calibration.conditioned_multitone` and measures the final conditioned qubit state reached from $|g, n\rangle$. The second, used as the primary optimization objective, measures the reduced effective qubit unitary on each manifold through a local helper that reuses the compiled multitone waveform but computes a two-level propagator sector by sector. For each scanned point we evaluated the baseline corrected pulse, an analytic small-angle warm start, reduced state-based optimization, and reduced effective-unitary optimization.

III. RESULTS

A. Convention Audit

The current `conditioned_multitone.py` source explicitly states that ϕ is the "rotation-axis angle" and recovers it from Bloch components through $\phi = \mathrm{atan2}(X, -Y)$. The earlier quarter-turn concern is therefore resolved in the present checkout, and the full study was rerun under that corrected interpretation.

B. Metric Comparison

Figure 2 shows that the reduced final-state objective is still weaker than the requested effective-unitary objective. Even after the convention fix, pulses optimized only to land close to the correct final state often deliver a noticeably poorer manifold-resolved unitary.

C. Duration and Active-Window Trends

The strongest nontrivial point occurred at $N_{\mathrm{active}} = 4$ and $|\chi|T/2\pi = 3.0$, where unitary optimization reached weighted process fidelity 0.999978 compared with a baseline value of 0.992316. The most informative improvement appeared in the shortest, hardest case: at $N_{\mathrm{active}} = 4$ and $|\chi|T/2\pi = 1.0$, the corrected baseline pulse achieved 0.793061, while unitary-based optimization raised that to 0.961990. Across the full grid, moderate duration was the sweet spot. The residual hard-case defect was a mixture of angle error and nonzero extracted axis-z tilt.

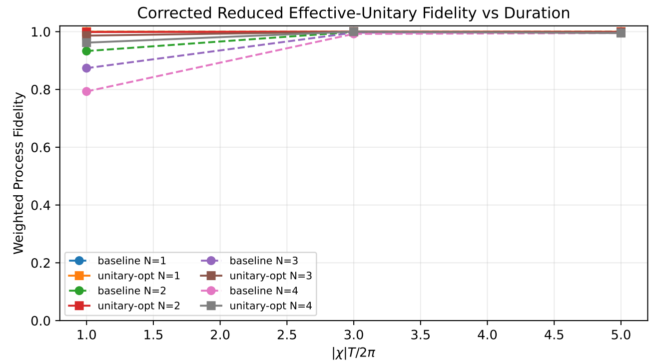

FIG. 1. Weighted reduced effective-unitary fidelity versus duration for the corrected direct multitone waveform before and after unitary-based optimization.

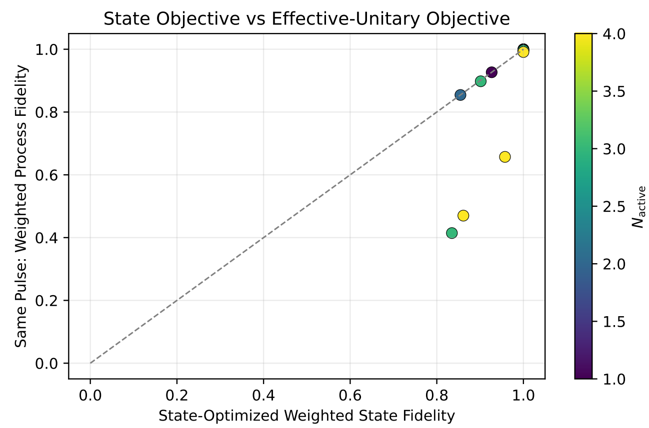

FIG. 2. Pulses optimized against the reduced final-state objective can still underperform on the stricter reduced effective-unitary metric.

D. Representative Four-Level Case

Figure 3 compares the target and achieved (θ_n, ϕ_n) values for the representative four-level case at $|\chi|T/2\pi = 3$. The optimized waveform tracks the intended parameters substantially better than the baseline while remaining visibly more faithful to the target than the state-optimized pulse.

IV. VALIDATION

A. Sanity Checks

The single-level case remained easy, with baseline reduced effective-unitary fidelity 1.000000. This matches the analytic expectation that one addressed sector carries no first-order multitone crosstalk.

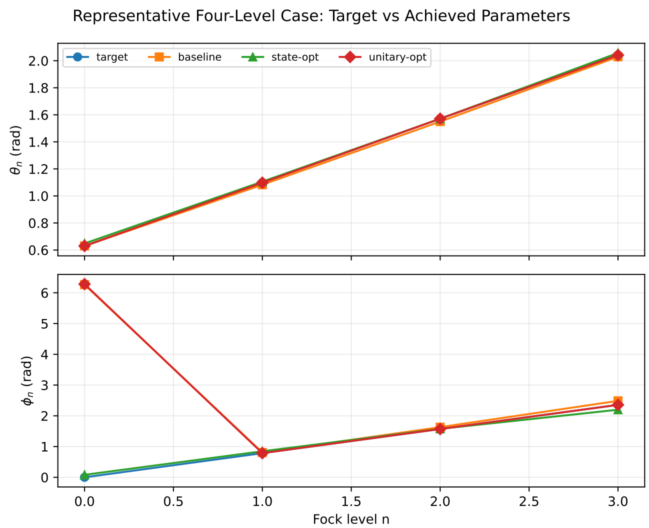

FIG. 3. Representative four-level case: target versus achieved effective rotation parameters after baseline, state-based optimization, and unitary-based optimization.

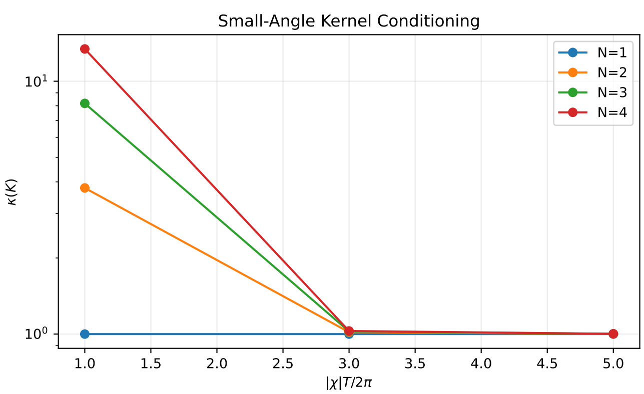

FIG. 4. Condition number of the small-angle multitone kernel. Full-rank kernels show that low-angle reachability exists in principle, but the growing condition number explains why the corrected direct ansatz becomes harder to tune as the active window expands.

B. Convergence Analysis

The best-case corrected solution was replayed at $dt = 8\,\mathrm{ns}, 4\,\mathrm{ns}, 2\,\mathrm{ns}$. The corresponding process fidelities were 0.999999, 1.000000, and 1.000000, indicating stable conclusions at the reported timestep.

C. Literature Comparison

No directly comparable published benchmark was available for this exact reduced effective-unitary metric, so the literature-comparison requirement is not applicable here.

V. DISCUSSION

The source-level phase fix matters because it removes an avoidable convention mismatch, but it does not change the deeper metric hierarchy. Once ϕ_n is interpreted consistently as a rotation-axis angle, the reduced final-state metric becomes a fairer proxy than before, yet it still remains weaker than the reduced effective-unitary objective requested by the user. The analytic kernel picture and the numerical optimization tell the same story: the corrected direct Gaussian multitone waveform is not obstructed in principle at low angle and is actually near ideal for this smooth target family at $|\chi|T/2\pi \approx 3$, even out to a four-level active window. The remaining challenge is concentrated in the shortest-duration cases, where multitone interference and residual axis-z tilt remain visible.

VI. CONCLUSION

The subtle quarter-turn issue in the reduced API has been fixed in the current `cqed_sim` checkout, and the study was rerun under that corrected convention. The rerun supports a positive but qualified conclusion: for the smooth corrected-SQR target used here, the direct Gaussian multitone family is expressive enough to realize near-ideal Fock-resolved qubit rotations at moderate duration, reaching 0.999978 weighted effective-unitary fidelity for a four-level active window at $|\chi|T/2\pi = 3$. The same family is less ideal in the shortest-duration regime, where even the best four-level point reached 0.961990. The most faithful optimization metric remains the reduced effective-unitary process fidelity, not the weaker final-state fidelity from $|g\rangle$ alone.

VII. LIMITATIONS AND FUTURE WORK

A. Known Limitations

The study used a closed-system reduced two-level model for the primary objective, scanned only three durations, and focused on one smooth four-level corrected-SQR target profile plus its smaller active subwindows.

B. Suggested Improvements

- **[P1 — MEDIUM]** Upstream a public reduced multitone effective-unitary extractor into `cqed_sim.calibration.conditioned_multitone`.
- **[P2 — MEDIUM]** Extend the rerun to sparse and random corrected-SQR targets.
- **[P2 — MEDIUM]** Replay the best reduced-unitary solutions with open-system noise.

C. Open Questions

The main unresolved question is whether a richer segmented or basis-modulated multitone waveform can exploit the now-correct phase convention to remove the residual axis-z tilt that remains under the direct Gaussian ansatz.

Appendix A: Detailed Results and Data

The machine-readable summary is stored in `data/study_summary.json` and the flattened comparison table in `data/case_table.csv`.

Appendix B: Reproducibility

1. Optimized Parameters

The literal highest-fidelity scan point was the trivial one-level case, for which the best corrections were

$$\delta\lambda = [0.0], \tag{B1}$$

$$\delta\alpha = [0.0], \tag{B2}$$

$$\delta\omega/2\pi \text{ (Hz)} = [0.0]. \tag{B3}$$

The representative four-level case highlighted in the main text used the corrections saved in `artifacts/representative_case_artifact.json`.

2. Waveform and Pulse Information

The representative waveform uses the corrected direct Gaussian multitone parameterization with shared duration $T = 1.056338e-06\,\mathrm{s}$ and the tone list saved in `artifacts/representative_case_artifact.json`.

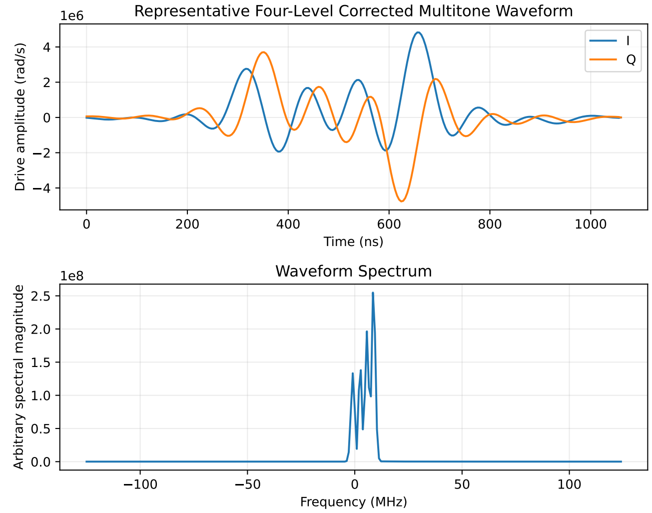

FIG. 5. Representative four-level corrected multitone waveform in time and frequency domains.

3. Gate Sequence and Decomposition

The study used a single direct multitone pulse with no appended cavity-only correction layer or echoed segment.

4. Modeling and Simulation Assumptions

All main-text results use the qubit-first dispersive model, no cavity self-Kerr, no open-system noise, a two-level qubit, and a reduced per-sector propagator objective.

5. Reproduction Procedure

1. Run `python scripts/run_study.py` from the study root.
2. Inspect `data/study_summary.json`, `data/validation_summary.json`, and `figures/`.
3. Compile `report/report.tex` with `pdflatex`, `bibtex`, `pdflatex`, `pdflatex`.

6. Saved Artifacts

- `data/study_summary.json`: full case-by-case study summary.
- `data/validation_summary.json`: sanity and convergence checks.
- `data/case_table.csv`: flattened numerical comparison table.

- `artifacts/best_case_artifact.json`: literal highest-fidelity case from the scan.
- `artifacts/best_case_waveform.npz`: sampled waveform arrays for the literal highest-fidelity case.
- `artifacts/representative_case_artifact.json`: representative four-level case highlighted in the report.
- `artifacts/representative_case_waveform.npz`: sampled waveform arrays for the representative four-level case.